

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2022/2023

JANUARY EXAMINATION

COMPUTER SCIENCE BACS2163
SOFTWARE ENGINEERING

MONDAY, 9 JANUARY 2023

TIME: 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS) IN DATA SCIENCE
BACHELOR OF COMPUTER SCIENCE (HONOURS) IN INTERACTIVE SOFTWARE
TECHNOLOGY

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS
DEVELOPMENT

BACHELOR OF SOFTWARE ENGINEERING (HONOURS)

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

BACS2163 SOFTWARE ENGINEERING**Question 1**

a) Answer the following questions based on Figure 1.

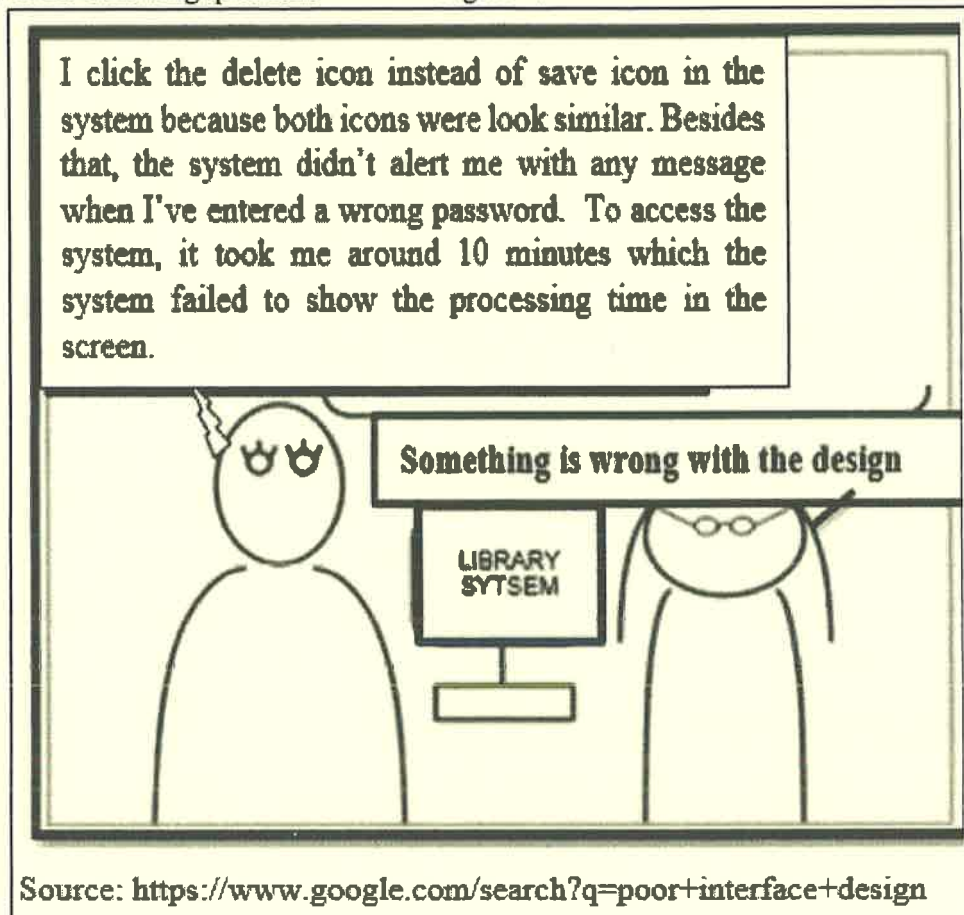


Figure1

- (i) Based on the above conversation in Figure 1, identify and explain the **THREE (3)** common design issues facing by user in the library system. (6 marks)
 - (ii) Based on answer in Question 1 a) (i), analyse how does this issue can be solved by using golden rules of user interface design. (9 marks)
- b) You, as an appointed software engineer, are in-charge for designing an online management system for A+ Tutor centre. Currently the tuition centre has been using manual way to manage centre and now has decided to automate all their registration, classes scheduling, attendance, and payment processes. Design 3 sets of aperiodic stimuli-response for A+ Tutor online management system as mentioned above. (6 marks)
- c) Differentiate between soft real time system and hard real time system with an example for each. (4 marks)
- [Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 2**

- a) You, as a project manager, have been assigned to lead project team to develop learning management system for ABC College. The main functions of the system include programme registration and payment, online assessment, assignment submission, uploading and downloading course materials, forum. The main campus of the college is located at Kuala Lumpur and the other branch campuses are located in Penang, Pahang, Johor, Perak and Sabah.

Recommend and illustrate an appropriate system organisation model for the above project with justification. (10 marks)

- b) Managing a project group is one of the challenging and critical management tasks in software development projects. Identify the **FOUR (4)** main responsibilities of a project manager. (8 marks)

- c) Compare and contrast between perfective maintenance and adaptive maintenance. Which maintenance is invoked in improving e-doctor application with login feature? Justify your answer. (7 marks)

[Total: 25 marks]

Question 3

- a) Mr Sam is using a net banking application to transfer the amount to his friend's account. So, he initiates the transaction, get a successful transaction message, and the amount also deducted from his account. However, his friend confirms that her account has not received any credits yet. Likewise, Mr Sam's account is also not reflecting the reversed transaction. This will surely make him upset and leave as an unsatisfied customer. Mr Sam claimed that it is because of lack of testing of the net banking application before release.

- (i) Do you agree with Mr. Sam's claim? Provide **THREE (3)** justification about the importance of software testing in your answer. (6 marks)

- (ii) Identify any **FIVE (5)** stages of software testing process. (10 marks)

BACS2163 SOFTWARE ENGINEERING**Question 3 (Continued)**

- b) Analyse the following Figure 2 and answer the questions below.

**Figure 2**

- (i) Identify any **TWO (2)** consequences for not having a test plan in their project. (4 marks)
- (ii) Outline the major **FIVE (5)** contents of the test plan. (5 marks)
- [Total: 25 marks]

Question 4

- a) Delta Company is developing a software for college library system. Among the requirements of the software is organising book systematically (author, year), stock checking and verification, search for book and its availability, purchase new books, checking out a book, returning a book and renewing a book. The development of the software is performed during the semester break period as the library staff may have more time to discuss in detail on the system. Delta has decided to apply viewpoint-oriented analysis in the requirement analysis stage.
- (i) Discuss **TWO (2)** advantages and **TWO (2)** disadvantages of using viewpoint-oriented analysis method in requirement gathering and analysis. (4 marks)
- (ii) Identify any **SIX (6)** viewpoints of college library system and prepare a viewpoint hierarchy. (12 marks)
- b) In a project management, a project manager would need to consider the factors that could motivate employees throughout the project. Suggest and explain **THREE (3)** motivating factors based on Maslow motivation model. (9 marks)
- [Total: 25 marks]